A Mobile Application for Inventory Management to Support Digital Transformation in Small Retail Businesses

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***Abstract*— Small retail businesses often rely on manual inventory management practices that lead to data inaccuracies and operational inefficiencies. This study presents Stock Easy, a lightweight mobile application designed to support inventory management and facilitate digital transformation among small retail businesses. The application was developed using a user-centered design approach and includes core functions such as product data management, stock tracking, and inventory updates. A quantitative evaluation was conducted with 132 users, including retail owners, online sellers, store managers, and inventory staff. Data were collected using a structured questionnaire and analyzed using descriptive statistics and Pearson correlation analysis. The results indicate high user satisfaction, particularly in terms of ease of use (mean = 4.52), process clarity (mean = 4.41), and user interface design (mean = 4.36). Correlation analysis further reveals that perceived ease of use and perceived usefulness significantly influence overall satisfaction, which is positively associated with perceived readiness for digital transformation. These findings demonstrate that lightweight digital tools such as Stock Easy can effectively improve inventory management practices and serve as an entry point for digital transformation in small retail businesses.**

Keywords*— *Digital Transformation, Inventory Management, Mobile Application, Small Retail Businesses, User Satisfaction, Technology Adoption

I. INTRODUCTION

Digital transformation (DT) has become a key driver of competitiveness and operational efficiency for businesses worldwide [1]-[2]. In the context of small and medium-sized enterprises (SMEs), the adoption of digital technologies can significantly improve productivity, decision-making, and supply chain management [3].

In Thailand, SMEs represent more than 99% of all enterprises and employ over 85% of the workforce, making them essential contributors to national economic development [4]. Despite their importance, many small retail businesses still rely on manual processes for operational management, particularly in inventory control [5]. Such traditional approaches often result in data inaccuracies, stock shortages, overstock situations, and inefficiencies in business operations [6]. Previous studies emphasize that digital transformation in SMEs should not necessarily begin with complex enterprise systems but rather with lightweight digital tools that align with the operational context and technological capabilities of small businesses [7]-[8]. Inventory management systems, for example, can improve data transparency, reduce operational errors, and support better planning and decision-making [8].

However, empirical studies focusing on user-centered digital tools tailored for small retail businesses in developing countries remain limited. Many existing systems are either too complex or costly for small retailers to adopt effectively. To address this gap, this study introduces Stock Easy, a mobile application designed to support inventory management and promote digital transformation among small retail businesses as in Fig. 1. The application focuses on essential functions that simplify stock monitoring and improve data management in everyday operations.

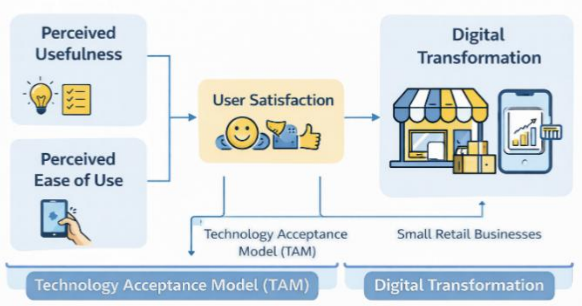

Fig. 1. Conceptual Framework (TAM + Digital Transformation).

The objectives of this study are: 1) To develop a mobile application (Stock Easy) for inventory management in small retail businesses. 2) To evaluate user satisfaction and usability of the application. 3) To investigate the role of the application in supporting digital transformation readiness. 4) To analyze the relationships between technology acceptance factors and user satisfaction.

II. RELATED WORKS

Digital transformation has emerged as a key driver of competitiveness and operational efficiency in small and medium-sized enterprises (SMEs), enabling improvements in productivity, decision-making, and value creation through technologies such as big data, IoT, and cloud systems [1]–[4]. Despite these advantages, SMEs often face constraints including limited resources, low digital literacy, and system complexity, which hinder effective technology adoption [3], [5]. Prior studies emphasize that user-friendly and lightweight digital solutions are more suitable for SMEs than complex enterprise systems, as they align better with operational needs and gradual adoption behaviors [6], [7]. In retail contexts, inefficient inventory management remains a critical issue, leading to data inaccuracies and increased operational costs, while mobile and cloud-based inventory systems have been shown to enhance data accuracy and efficiency [8], [9]. From a behavioral perspective, the Technology Acceptance Model (TAM) explains that

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perceived ease of use and perceived usefulness significantly influence user satisfaction and continued system usage [5], [10]. However, existing research has largely examined digital transformation, inventory systems, and technology acceptance independently. Limited studies integrate these perspectives within the context of mobile inventory management applications tailored for small retail businesses, particularly in developing economies. This study addresses this gap by investigating how a lightweight mobile application can simultaneously improve operational efficiency and support digital transformation readiness among SMEs.

III. METHODOLOGY

A. System Development

The Stock Easy application was developed using a user-centered design approach to ensure usability for small retail business operators with varying levels of digital literacy as show in Fig. 2.

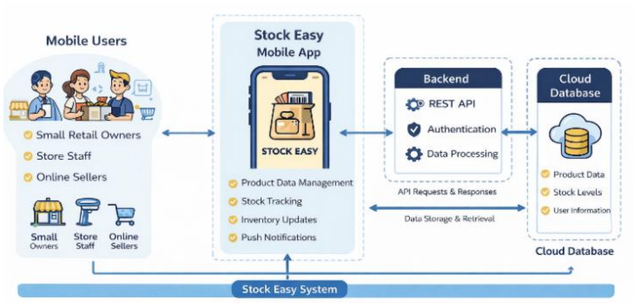

Fig. 2. Stock Easy System Architecture Diagram.

The system supports the following core functions:

- Product data management (product ID, product name, stock quantity)
- Expiration date tracking
- Product image storage
- Product search functionality
- Add, edit, and delete inventory records

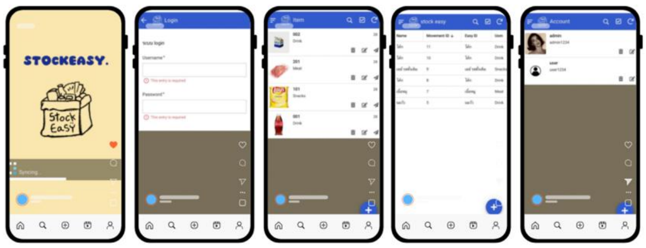

Fig. 3. User interface (UI) of the mobile application for inventory management.

These functions aim to simplify inventory management tasks while reducing reliance on manual record-keeping.

B. Participants

The study involved 132 participants who had actual experience using the Stock Easy application. The participants were categorized into four groups:

- Small retail business owners (31)
- Online sellers (27)
- Shop owners or store managers (41)
- Inventory and frontline staff (33)

Participants were selected using purposive sampling to ensure relevant user experience.

C. Data Collection Instrument

The measurement items for key constructs, including Perceived Ease of Use (PEOU) and Perceived Usefulness (PU), were adapted from the original Technology Acceptance Model (TAM) proposed by Davis (1989) and subsequent validated studies in SME and mobile application contexts. Minor wording adjustments were made to align with the inventory management scenario. All items were reviewed by three domain experts to ensure contextual relevance prior to pilot testing.

IV. DATA ANALYSIS

The data analysis was conducted in three stages to examine the usability of the application and the relationships among key variables.

A. Construct Measurement

Each construct was measured using multiple questionnaire items. The composite score of each construct was calculated as the average of its associated items:

$$X_k = \frac{1}{n_k} \sum_{i=1}^{n_k} x_{ki} \tag{1}$$

where X_k represents the composite score of construct k, x_{ki} denotes the score of item i for construct k, and n_k represents the number of items measuring the construct.

The main constructs used in this study include:

- Perceived Ease of Use (PEOU)
- Perceived Usefulness (PU)
- Process Clarity (PC)
- User Satisfaction (US)
- Digital Transformation Readiness (DTR)

B. Research Model

Based on the Technology Acceptance Model (TAM) and digital transformation theory, user satisfaction is modeled as a function of perceived ease of use, perceived usefulness, and process clarity as in Fig 4. The relationships among variables are defined as follows:

$$US = \beta_0 + \beta_1 PEOU + \beta_2 PU + \beta_3 PC + \varepsilon_1 \tag{2}$$

where US represents user satisfaction, $PEOU$ represents perceived ease of use, PU represents perceived usefulness, and PC represents process clarity.

Furthermore, digital transformation readiness is modeled as a consequence of user satisfaction:

$$DTR = \gamma_0 + \gamma_1 US + \varepsilon_2 \tag{3}$$

where DTR represents digital transformation readiness.

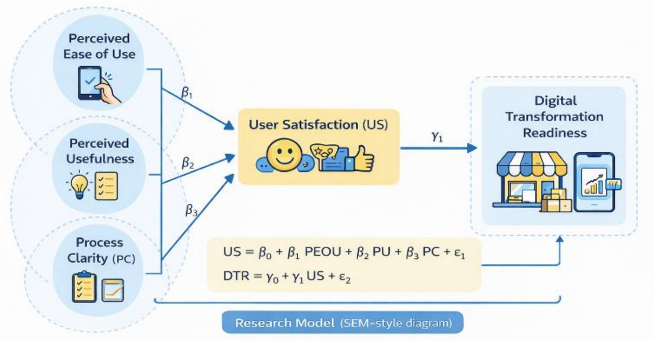

Fig. 4. Research model illustrating the relationships among perceived ease of use, perceived usefulness, process clarity, user satisfaction, and digital transformation readiness.

C. Correlation Analysis

To examine the relationships among variables, Pearson's correlation coefficient was used. The correlation coefficient is calculated as:

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2 \sum_{i=1}^{n}(y_i-\bar{y})^2}} \quad (4)$$

where r_{xy} represents the correlation coefficient between variables x and y, x_i and y_i are individual observations, and $\bar{x}$ and $\bar{y}$ denote the sample means.

Statistical significance was evaluated at $p < 0.05$ and $p < 0.01$.

V. Experimental Setup and Results

A. Experimental Setup

Before completing the questionnaire, participants were asked to use the Stock Easy application in their daily inventory management tasks over a two-week trial period. This duration was considered sufficient to allow users to become familiar with the system's core functionalities and integrate it into routine operations. Prior to the trial, a short onboarding session (approximately 30 minutes) was conducted to introduce the system features and ensure consistent understanding among participants. After the trial period, respondents completed the structured questionnaire to evaluate usability, satisfaction, and perceived impact on digital transformation readiness.

TABLE I. User Satisfaction Evaluation Results (N = 132)

Variable	Category	Percentage (%)
Gender	Male	42.3
	Female	57.7
Age	18–24	48.6
	25–34	32.4
	35+	19.0
Business Experience	< 3 years	38.2
	3–5 years	34.6
	> 5 years	27.2
Digital Experience	High	44.1
	Moderate	39.5
	Low	16.4

B. User Satisfaction Results

Table I presents the descriptive statistics of user satisfaction toward the mobile inventory management application.

TABLE II. User Satisfaction Evaluation Results (N = 132)

Evaluation Dimension	Mean	S.D.	Satisfaction Level
Ease of Use	4.52	0.51	Very High
Process Clarity	4.41	0.56	High
UI Design	4.36	0.58	High
Overall Satisfaction	4.43	0.49	High

The results indicate that users expressed high overall satisfaction with the application. Among all dimensions, ease of use obtained the highest mean score (4.52), suggesting that the application interface and workflow were easy to understand even for users with different levels of digital literacy. Process clarity and user interface design also received high ratings, demonstrating that the application successfully supports practical inventory management tasks in small retail businesses.

C. Role in Supporting Digital Transformation

Table III presents users' perceptions of the system's contribution to digital transformation in small retail operations.

TABLE III. Perceived Impact on Digital Transformation

Evaluation Item	Mean	S.D.	Level
Reduces inventory data errors	4.47	0.53	High
Reduces redundant recording tasks	4.39	0.55	High
Improves data availability for decision-making	4.42	0.52	High
Supports digital transformation readiness	4.45	0.50	High

The findings show that users perceived the application as an effective tool for reducing stock management errors and improving operational efficiency. The application also enhanced data availability for business decision-making, which is an important factor for digital transformation in small retail environments.

TABLE IV. CORRELATION MATRIX

Variables	PEOU	PU	PC	US	DTR
PEOU	1.00	0.68**	0.64**	0.72**	0.61**
PU	0.68**	1.00	0.66**	0.74**	0.69**
PC	0.64**	0.66**	1.00	0.70**	0.63**
US	0.72**	0.74**	0.70**	1.00	0.76**
DTR	0.61**	0.69**	0.63**	0.76**	1.00

Note: $p < 0.01$

The correlation analysis reveals strong positive relationships among all key constructs. In particular, user satisfaction shows a strong correlation with digital transformation readiness (r = 0.76), confirming its critical role as a mediating factor in SME digital adoption.

D. Visualization of User Satisfaction

The bar chart in Fig. 5 illustrates the average satisfaction scores across the four evaluation dimensions.

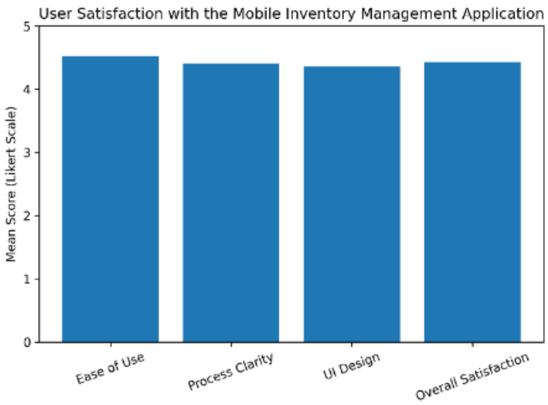

Fig. 5. Bar chart illustrating user satisfaction levels across key usability dimensions of the mobile inventory management application.

The visualization confirms that all usability dimensions achieved mean scores above 4.3, indicating strong acceptance of the system among participants. Ease of use obtained the highest rating, reinforcing the importance of simple and intuitive design when developing digital tools for small retail businesses.

VI. Conclusion

This study demonstrates that a lightweight mobile application can significantly enhance inventory management efficiency and support digital transformation in small retail businesses. The findings confirm that perceived ease of use, usefulness, and process clarity are key determinants of user satisfaction, which in turn drives digital transformation readiness. The results highlight the importance of simple, user-centered digital solutions as an effective entry point for SMEs transitioning from manual to digital operations.

VII. Discussion

The results of this study provide several insights into the role of mobile applications in supporting digital transformation among small retail businesses. The findings indicate that users reported high satisfaction with the proposed inventory management application, particularly in terms of ease of use, process clarity, and user interface design. These results suggest that simple and intuitive digital tools can effectively support operational tasks in small retail environments. One important finding of this study is the strong influence of perceived ease of use on overall user satisfaction. This result aligns with the Technology Acceptance Model (TAM), which identifies ease of use as a key determinant of technology adoption. When users perceive a system as easy to learn and operate, they are more likely to develop positive attitudes toward its continued use. In the context of small retail businesses, where users may have varying levels of digital literacy, usability becomes a critical factor in determining whether a digital tool can be successfully adopted. The findings also highlight the importance of perceived usefulness in shaping user satisfaction. Participants indicated that the application helped reduce inventory errors, minimized redundant manual recording, and improved access to product information. These practical benefits reinforce the idea that digital transformation in small businesses often begins with improvements in everyday operational processes rather than large-scale technological changes.

Another notable result is the role of process clarity in influencing user satisfaction. Clear workflows and well-structured interfaces allow users to complete tasks efficiently without confusion. This is particularly important for small retail operations, where inventory management tasks must be performed quickly and accurately during daily business activities.

Furthermore, the results demonstrate that user satisfaction is positively associated with digital transformation readiness. When users experience positive outcomes from using digital tools, they become more open to adopting additional digital technologies in their operations. This finding supports the concept of incremental digital transformation, where small-scale digital solutions gradually build organizational readiness for broader technological adoption.

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